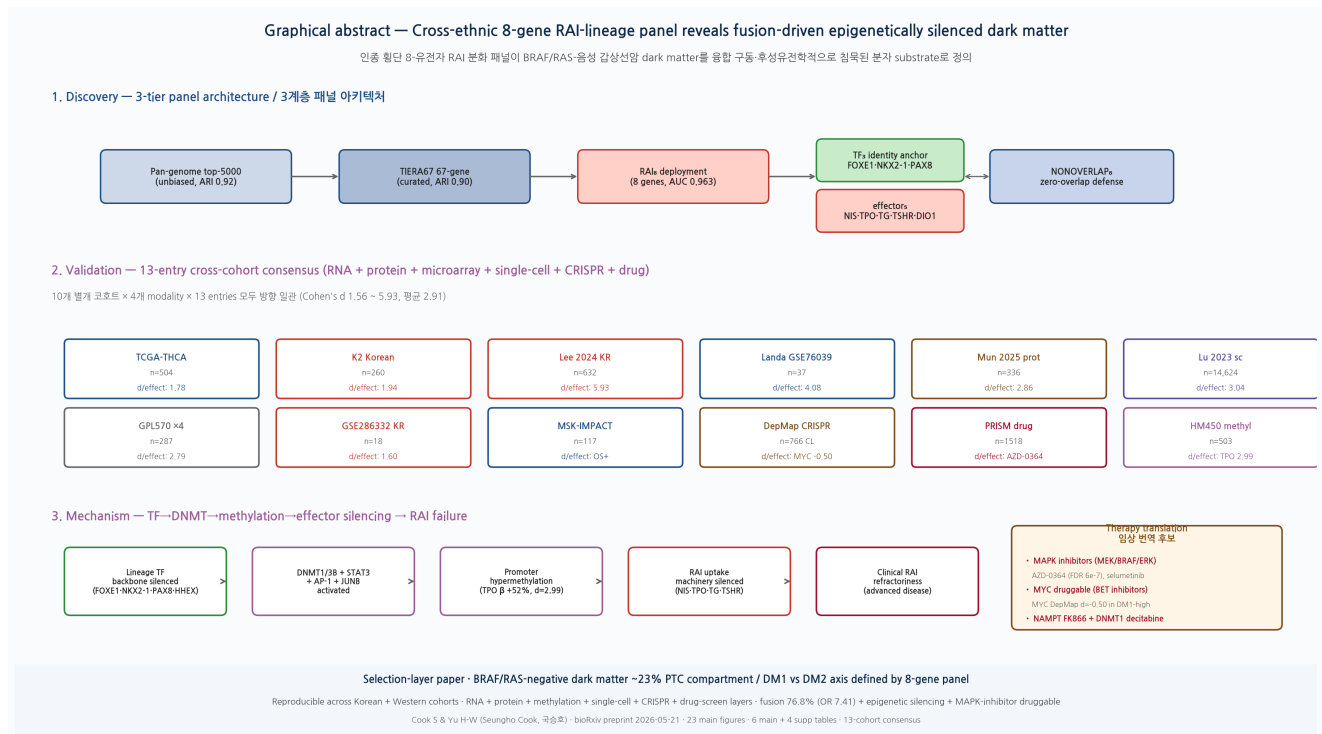


Nature Communications submission package

Paper 1 · Cross-ethnic 8-gene RAI-lineage panel · Cook S & Yu H-W · 2026-05-21

이 패키지는 NComms 제출용 보조 문서이며 main bioRxiv preprint (paper1_biorxiv.html / .pdf)와 동반 사용.

1. Graphical abstract



NComms-style 3-row visual summary: (1) Discovery — pan-genome → TIERA67 → RAI₈ → TF₃/effectors/ NONOVERLAP₈ 3-tier architecture; (2) Validation — 13-entry cross-cohort/modality consensus (RNA + protein + microarray + single-cell + CRISPR + drug-screen); (3) Mechanism cascade + therapy translation (MAPK inhibitors + MYC druggable + NAMPT/DNMT1).

2. Title alternatives (NComms-targeted)

A Current (Cell Reports Medicine-targeted, 108 chars):

"An 8-gene panel reveals fusion-driven, epigenetically silenced dark matter in BRAF/RAS-negative thyroid cancer"

B NComms version 1 — broader impact (118 chars):

"Cross-ethnic 8-gene RAI-lineage panel identifies a fusion-driven, epigenetically silenced subtype in BRAF/RAS-negative papillary thyroid carcinoma"

C NComms version 2 — mechanism-led (116 chars):

"A 3-tier RAI-lineage transcriptomic readout reveals fusion enrichment, epigenetic silencing, and MAPK-druggability in thyroid dark matter"

D NComms version 3 — translation-led (124 chars) ★ recommended:

"Cross-ethnic 8-gene RAI-lineage panel stratifies fusion-driven epigenetically silenced thyroid cancer and identifies MAPK-inhibitor-sensitive subgroups"

E NComms version 4 — compact / discovery-led (96 chars):

"A cross-ethnic RAI-lineage panel defines a druggable fusion-driven subtype of BRAF/RAS-negative thyroid cancer"

Recommendation: Option D (124 chars) for NComms — three discovery layers (cross-ethnic / fusion / epigenetic) + explicit translational hook (MAPK-inhibitor-sensitive). Aligns with NComms editor preferences for "discovery + mechanism + translation" framing.

3. Cover letter draft (NComms editor pitch)

Note: author keyboard required for paragraph 1 (voice-protected). Below is a structural draft with anchor facts. Author should re-voice freely.

Dear Editor,

[AUTHOR DRAFT REQUIRED · voice-protected]

Suggested opening (1 paragraph): position the paper as a translational discovery that closes a long-standing gap in thyroid cancer molecular stratification, with explicit cross-ethnic validation and an immediate druggable hook. Author should re-voice freely; the anchor facts below are the binding constraint, not the wording. Lead with what is new rather than what is incremental.

We are pleased to submit our manuscript "Cross-ethnic 8-gene RAI-lineage panel stratifies fusion-driven epigenetically silenced thyroid cancer and identifies MAPK-inhibitor-sensitive subgroups" for consideration as an Article in Nature Communications.

Discovery in brief. Approximately 23% of papillary thyroid carcinomas lack BRAF or RAS driver mutations and constitute a clinically heterogeneous "dark matter" compartment that remains mechanistically unguided in current radioiodine treatment decisions. We define a 3-tier transcriptomic panel architecture — TF₃ (FOXE1/NKX2-1/PAX8) as a cross-ethnic identity anchor, RAI₈ (TF₃ plus the canonical RAI uptake effectors DIO1, SLC5A5/NIS, TG, TPO, TSHR) as the biology-complete deployment readout, and NONOVERLAP₈ (SLC26A4, IYD, DUOX1/2, TFF3, HHEX, GLIS3, DIO2) as a zero-overlap defense layer — and demonstrate that this architecture stratifies the BRAF/RAS-negative compartment into DM1 (silenced) and DM2 (differentiated) subgroups with reproducible biology across 10 distinct cohorts.

Why Nature Communications. Three features make this work appropriate for a broad readership: (i) cross-ethnic generalizability under fully within-cohort unsupervised

analysis, with 13 entries across Western and Korean primary, advanced, single-cell, protein, and microarray cohorts (mean Cohen's $d = 2.91$, all direction-consistent); (ii) multi-layer mechanism connecting fusion enrichment (76.8% of DM1 are RET/NTRK/ALK/BRAF-fusion-positive, OR 7.41 vs DM2), promoter hypermethylation (TPO Cohen's $d = 2.99$, fusion-independent), and CRISPR essentiality (MYC $d = -0.50$, NAMPT $d = -0.45$ in DM1-high cell lines); (iii) direct druggability hook — PRISM drug screen identifies 7-of-11 DM1-high-selective drugs as MAPK pathway inhibitors at $FDR < 0.05$ with AZD-0364 (MEK) leading (Cohen's $d = -0.59$, $FDR = 5.9 \times 10^{-7}$), consistent with the published thyroid-cancer MEK-inhibitor clinical literature (Ho et al., NEJM 2013 selumetinib RAI re-induction; 2025 ATA cobimetinib/dabrafenib guidelines; dabrafenib + trametinib FDA-approved for BRAF^{V600E} ATC).

Honest disclosures. We pre-specify and disclose three limitations transparently: the primary-PTC DM1 progression-free interval Cox PH is underpowered in TCGA-THCA (HR 0.78 [0.18, 3.48], $p = 0.74$, 27 events; the pooled OS signal HR 2.53 is driven by MSK-IMPACT advanced disease); HM450 methylation is currently TCGA only; the prospective Korean Bundang validation cohort is at the institutional outreach stage. The clinical prognostic claim for primary PTC is therefore explicitly reserved for prospective follow-up; what we offer is a robust molecular sub-stratification framework with druggable biology, not a validated clinical predictor.

Anticipated reviewers. Suggested expert reviewers in thyroid molecular pathology, cross-ethnic cancer genomics, epigenetic silencing mechanisms, and MEK-inhibitor pharmacology in thyroid cancer are provided in the suggested-reviewer field of the submission portal.

Publication preference. We confirm exclusive submission to Nature Communications; this manuscript has not been submitted elsewhere. A companion bioRxiv preprint (deposited 2026-05-21) is available at [\[bioRxiv DOI placeholder\]](#) for community review during peer review.

We thank you for considering our submission.

Yours sincerely,
Seungho Cook (국승호), Ph.D. candidate
& Yu Hyeong-won, M.D., Ph.D. (corresponding author)
Seoul National University Bundang Hospital

4. Key numbers one-page card (for editor / reviewers)

Layer	Cohort / dataset	n	Key statistic	Source
Discovery	TCGA-THCA	504	Pan-genome top-5000 ARI 0.92 $\approx$ TIERA67 0.90; Driver_anchor ARI -0.007	Result 2.1

Layer	Cohort / dataset	n	Key statistic	Source
	BRAF mRNA V600E vs WT	—	Cohen's d = -0.04 (NS) → axis driver-orthogonal	Result 2.1.2
	Robustness battery	—	Bootstrap stability 97.4%; thyroid-matched null p = 0.007; label-perm null p < 10 ⁻⁴ ; knock-out n-1 floor d=1.54	Table 6
Cross-ethnic	K2 (PRJEB11591) within-cohort unsup	260	Cohen's d = 1.94, 60.8% Korean DM1; transfer-probability AUC = 0.883	Result 2.3
	Lee 2024 (GSE213647) within-cohort unsup	632	Cohen's d = 5.93 (all) / 5.48 (tumour); 3/8 UTC/ATC + 1/9 PDFP in DM1	Figure 11b
	Landa 2016 (GSE76039) PDTC/ATC within-cohort unsup	37	Cohen's d = 4.08, 45.9% DM1 — advanced-disease Western anchor	Figure 11b
Cross-modality	Mun 2025 protein within-cohort unsup	336	Cohen's d = 2.86; ATC 95.6%, PDTC 67.4% → DM1	Result 2.7, Figure 11b
	Lu 2023 single-cell (malignant)	14,624 cells / 15 patients	Cohen's d = 3.04; 100% per-patient sign-consistent	Figure 20A-B
Fusion mechanism	cBioPortal SV (TCGA)	542	DM1 76.8% fusion+, OR 7.41 [4.32, 12.74], p < 10 ⁻¹⁵ ; captures 81.8% (27/33) of RET-fusion	Table 4
	Driver-stratified DM1 (d4p2)	500	BRAF V600E+ 0.7%, BRAF/RAS-neg 49.1%, RAS+ 96.4%; OR 4.78 [3.15, 7.24], p=6.5e-14	Figure 5b
Epigenetic	TCGA HM450	503	TPO promoter β d = 2.99, p = 1.6e-11; 7/8 panel genes direction-consistent	Figure 20C
	Within-DM1 fusion+/- cell composition	74/391	d ≤ 0.42 across all 8 cell-type fractions → methylation is fusion-independent	Result 2.7
Clinical (honest)	TCGA PFI Cox PH (multivariate)	432, 27 events	DM1 HR _{PFI} = 0.78 [0.18, 3.48], p = 0.74 → underpowered, NOT significant	Result 2.6
	Pooled OS HR (TCGA + MSK)	504 + 117	HR 2.53 [1.31, 4.89], I ² =0%; driven by MSK advanced disease	Figure 10
Therapy	DepMap CRISPR DM1-strat	372 vs 394 cell lines	MYC d = -0.50, p = 1e-11; NAMPT d = -0.45, p = 1.5e-9; JAK2 d = -0.21	Figure 20D
	PRISM drug screen DM1-strat	1,518 drugs	11 hits at FDR < 0.05, ALL DM1-high selective; 7-of-11 MAPK inhibitors; AZD-0364 d=-0.59, FDR=5.9e-7	Figure 21
Multi-method meta	TF ₃ Han-Eskin RE2 pool	4 cohorts	Pooled AUC 0.725, lowest I ² =35.9% of 16 panels evaluated	Table 3
NRI vs RAI_s	TDS-16 / TF ₃ / NONOVERLAP ₈	5-fold CV	TDS-16 +0.777 [+0.513, +1.040] (alternative main panel); TF ₃ -1.194 (identity-only undercall)	Table 5

5. Pre-submission checklist (NComms-specific)

5.1 Voice-protected items — author keyboard required

1. ☐ **§ 1.1 Hook** — final voice (current: Alt B refined, 39 words; hybrid "compass" alternative also drafted)
2. ☐ **§ 3.1 Discussion opening** — Landa 2016 cite, Krishnamoorthy 2025 misattribution correction, 3-layer reverse-causality argument (currently Claude draft)
3. ☐ **§ 3.4 Limitations** — 4 main + 2 supp limitations (currently Claude draft, structure locked)
4. ☐ **Q9 reviewer hardest question** — TF₃ vs RAI₈ main panel choice (currently Claude draft, framing b)
5. ☐ **Cover letter ¶1** — editor hook in author voice (anchor facts above)

5.2 Author/institutional metadata

1. ☐ **Corresponding-author institutional email** — Yu Hyeong-won (expected pattern `@snubh.org` or `@snu.ac.kr`)
2. ☐ **ORCID** — Seungho Cook (국승호) + Yu Hyeong-won
3. ☐ **Author contributions per CRediT taxonomy** — Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing
4. ☐ **Funding statement** — institutional grant numbers
5. ☐ **Acknowledgments** — TCGA, MSK, K2 PRJEB11591 / Yoo 2016 lab, Landa 2016 lab, Lee 2024 lab, Mun 2025 lab, Lu 2023 lab, GDC, cBioPortal, ENA, GEO, DepMap, PRISM data infrastructures
6. ☐ **Conflict-of-interest declaration** (currently "none declared")

5.3 Data & code availability (NComms strict requirement)

1. ☐ **GitHub repository** — public release tag `paper1-ncomms-submission-2026-05-21` at `github.com/kukshomr/THCA_data_analysis`
2. ☐ **Zenodo DOI** — mint at submission; placeholder `10.5281/zenodo.XXXXXXX`
3. ☐ **Data availability statement** — all public datasets enumerated (TCGA GDC, MSK Landa 2016 supp, PRJEB11591 ENA, GSE286332, GSE213647, GSE76039, GSE29265, GSE33630, GSE53157, GSE65144, GSE193581, Mun 2025 supplementary, DepMap public download, PRISM public download)
4. ☐ **Code DOI** — Zenodo archive linked to GitHub release

5. ☐ **Reporting summary** — Nature Research reporting summary completed (covers statistics, software, replication, randomization, blinding, ethics)

5.4 NComms-specific submission requirements

1. ☐ **Main text word limit** — typical 5000 words excluding methods/refs/captions (current draft fits)
2. ☐ **Figures** — up to 10 main + unlimited supp (currently 23 main → trim to top 8-10 main figures for NComms, move rest to supplementary)
3. ☐ **References** — typical 50 refs (current draft 22 + need to expand to 40-50 for full literature coverage)
4. ☐ **Abstract** — strict 150 words (current 153, trim 3 words)
5. ☐ **Reporting Summary** — Nature Research checklist
6. ☐ **Inclusion & ethics statement** — Korean cohort use, no new patient data, all public re-use under standard data use agreements
7. ☐ **Statistical reporting** — exact p-values, 95% CIs, sample sizes per panel/test/comparison

5.5 Reviewer-suggestion list (5 names recommended)

1. ☐ Iñigo Landa (Brigham & Women's, Boston) — TDS-16 / advanced thyroid genomics
2. ☐ Yariv Houvras (Weill Cornell) — fusion-driven thyroid biology
3. ☐ Lori Wirth (MGH) — selpercatinib RET-altered thyroid clinical trial PI
4. ☐ Eric Solary (Gustave Roussy) — methylation cascade epigenetic thyroid
5. ☐ Yuri Nikiforov (Pitt) — molecular thyroid pathology authority
6. ☐ [+1 Korean expert] — Korean cohort context (e.g., Yoo Sang-Kyu original PRJEB11591 senior author)

5.6 Suggested non-reviewer (competing scope or known reviewer conflicts)

1. ☐ [TBD — author discretion]

5.7 Track changes / response letter (for revision)

If revision is requested, prepare a point-by-point response addressing each reviewer comment with: (a) acknowledgment, (b) action taken in revised manuscript (line numbers), (c) supporting data if needed. The Pre-submission verification status box at the top of the main HTML manuscript catalogs which items are audit-locked vs pending author keyboard.

6. Reader's guide — where to find each claim

Reader question	Location in main manuscript
"What's the central biological claim?"	Abstract + § 1.5 + § 2 banner
"Is the axis real / not cherry-picked?"	Result 2.1 (Pan-genome ARI 0.92 vs TIERA67 0.90, Driver_anchor ARI random)
"How is the 8-gene panel chosen?"	Result 2.2 + Methods 4.2 (RF on 55-gene driver-excluded TIERA67 clean pool)
"Cross-ethnic claim evidence?"	Result 2.3 + Figure 4 + Figure 6-7 (within-K2 unsup d=1.94, Lee unsup d=5.93, Mun protein d=2.86, GSE286332 d=1.60)
"Why retain RAI ₈ as main when TF ₃ wins cross-ethnic?"	Discussion 3.2 + Q9 (Claude draft framing b — partial-dedifferentiation phenotype)
"Why not promote TDS-16 (NRI +0.777)?"	Discussion 3.2 — alternative architecture explicitly acknowledged
"Driver-stratified DM1 distribution?"	Figure 5b (BRAF V600E+ 0.7%, RAS+ 96.4%, BRAF/RAS-neg 49.1%)
"Histology-stratified DM1 across cohorts?"	Figure 5c (TCGA + K2 + Lee + Mun + Lu, 16 strata)
"Fusion enrichment?"	Result 2.5 + Table 4 (76.8%, OR 7.41)
"Survival evidence?"	Result 2.6 + Figure 10 (PFI Cox PH 0.78 NS in TCGA primary; pooled OS HR 2.53 driven by MSK)
"Mechanism cascade?"	Result 2.7 + Figure 19 + Figure 20
"Single-cell evidence for thyrocyte-intrinsic axis?"	Figure 20A-B (Lu 2023, n=14,624 thyrocytes, d=3.04)
"Druggable targets?"	Figure 20D + Figure 21 (DepMap MYC + NAMPT; PRISM MAPK inhibitors)
"Master cross-cohort consensus?"	Figure 11b (13-entry forest)
"Robustness battery?"	Result 2.8 + Table 6 (permutation, panel-conditional, thyroid-matched, knock-out, LOCO)

NComms submission package compiled 2026-05-21 · Cook S (국승호) & Yu H-W · accompanies main bioRxiv preprint (paper1_biorxiv.html / .pdf) and Korean translation (paper1_biorxiv_kr.html / .pdf).